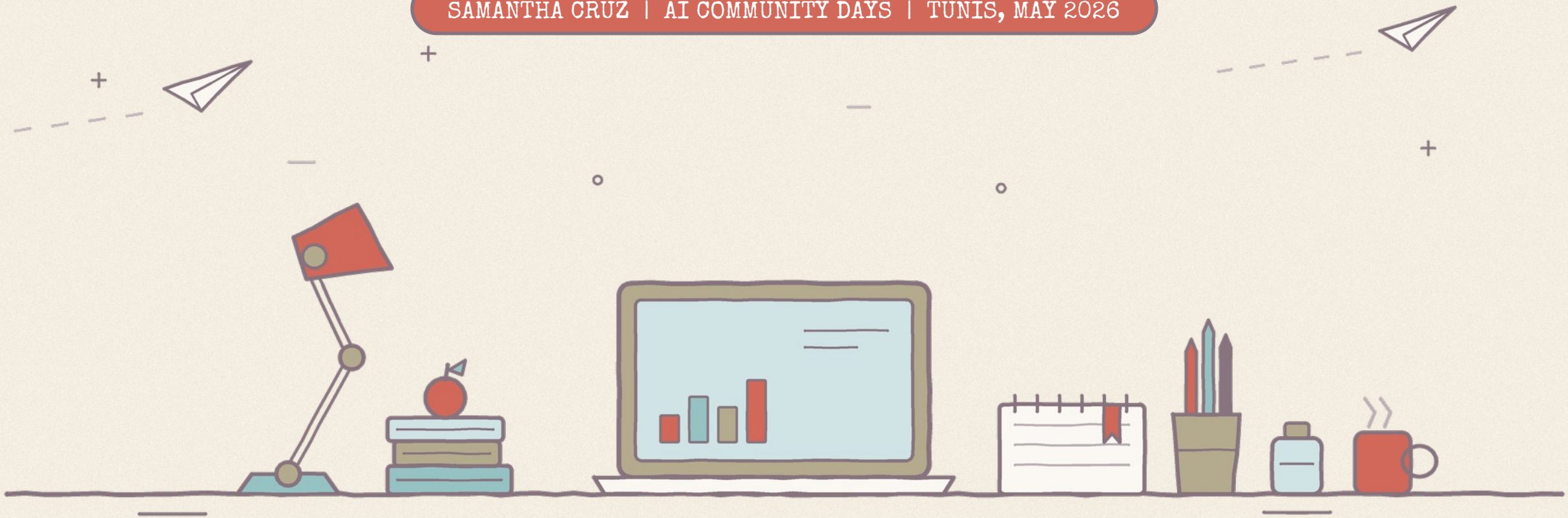
AI COMMUNITY DAYS

HOW MUCH DOES A PROMPT COST?

Under the hood of AI agents, Copilot and Microsoft Fabric capacity

SAMANTHA CRUZ | AI COMMUNITY DAYS | TUNIS, MAY 2026



HI, I'M SAMANTHA

DATA ENGINEERING TEAM LEAD · BARCELONA

- **Team lead at isolutions:** lakehouse platforms on Fabric, Databricks and Azure
- **Databricks instructor** since 2024
- **Started in oceanography:** satellite and ocean data, then BI and data engineering
- **Past teams:** Glovo, AB InBev, APCOA
- **Speaker and writer:** community events and Medium

8+ years in data

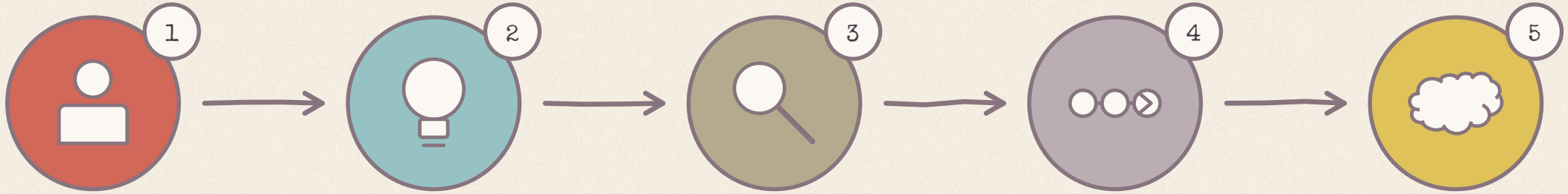
4 Fabric and Databricks certifications





A PROMPT IS NEVER JUST A PROMPT

FIVE SYSTEMS RUN BEFORE YOU SEE AN ANSWER



Prompt

Your question plus instructions and history

Tokens

The model reads and writes in tokens

Query

Tools and data questions run real queries

Pipelines

Fresh, prepared data had to be built first

Capacity

Fabric capacity pays for all of it

TAKEAWAY **The visible cost is the model. The hidden cost is everything around it.**

TOKENS ARE THE METER

WHAT THE MODEL BILLS YOU FOR

Input

- Your question
- System instructions
- Retrieved data

History

- Resent on every turn
- Grows with the chat
- Quietly multiplies cost

Output

- The generated answer
- Usually priced higher
- Longer means costlier

Tool calls

- Each step re-reads context
- Agents loop and retry
- Cost per question varies

TAKEAWAY **Cost follows context size, not question length.**

SAME TOOL, THREE VERY DIFFERENT BILLS

QUALITATIVE COMPARISON OF THREE SCENARIOS

SCENARIO	WHAT RUNS	MAIN COST DRIVERS	RELATIVE COST
Simple prompt	One model call	Input and output tokens	Low
Data question	Model, generated query, answer	Tokens plus query compute	Medium
Agent workflow	Many calls, tools and retries	Tokens, queries, pipelines	High and variable





WHERE THE COST ACCUMULATES

FOUR LAYERS, ONE SHARED BILL

CAPACITY	**Fabric capacity units** Shared pool that Copilot, reports and jobs all draw from
PIPELINES	**Data preparation** Refreshes and transformations behind the answer
QUERIES	**Query execution** SQL, DAX or Spark work done on the prompt's behalf
MODEL	**Token usage** Input, output and context for every call

REMEMBER

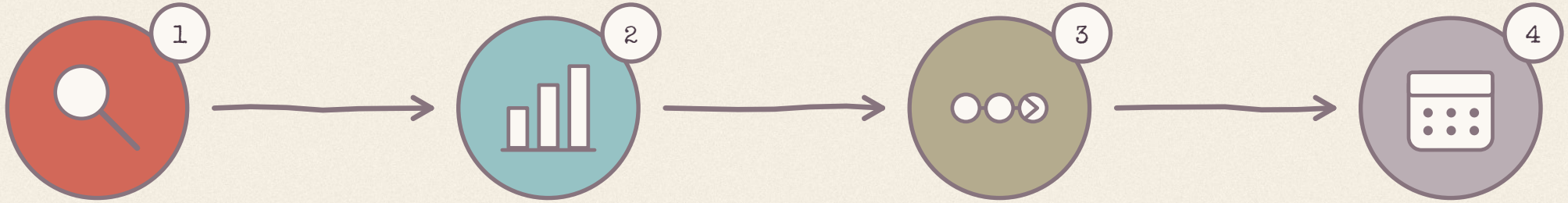
A busy AI assistant can slow down your reports.





FROM USAGE TO REAL MONEY

A BACK-OF-THE-ENVELOPE METHOD



Measure

Tokens and queries for one typical prompt

Price

Multiply by token price and capacity cost

Add extras

Include pipelines and refreshes behind it

Scale

Users × prompts per day × working days

TAKEAWAY **A cheap prompt times a big team is a big number.**

WHERE TEAMS UNDERESTIMATE

COMMON MISCONCEPTIONS VS REALITY

WHAT WE ASSUME

- ✗ Cost equals the model price
- ✗ Every prompt costs the same
- ✗ Pilots predict production
- ✗ Capacity is free once bought

WHAT REALLY HAPPENS

- ✓ Queries and pipelines add up
- ✓ Agents loop, so cost varies
- ✓ Scale and history multiply cost
- ✓ Everything competes for capacity

TAKEAWAY **Estimate the whole chain, not just the model.**





DESIGN FOR PREDICTABLE COST

SIX HABITS FOR PRODUCTION AI

Trim context

- Send only what is needed
- Summarise long history

Right-size models

- Small model for easy tasks
- Big model only when needed

Pre-aggregate

- Curated gold tables
- Cheaper, faster queries

Cap agent loops

- Limit steps and retries
- Fail early and clearly

Cache answers

- Reuse frequent results
- Skip repeated queries

Monitor usage

- Track cost per feature
- Alert on spikes

THANK YOU

Measure the whole chain, not just the model
Design for context, queries and capacity
Turn usage into money before production does

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